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Project in machine learning at MATF

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§ 1 The pitch

A guitar amplifier is a nonlinear system with memory. It clips and compresses the signal in a way that depends on how hard it is driven, and the speaker cabinet that follows colours the result with a long impulse response. Modelling one by hand means measuring circuits; modelling one as a black box means learning the mapping from a dry recording to the amplified one.

That is what this project does. Given a pair of recordings – the same performance clean and processed – we train a model to reproduce the processing, so that the amplifier can be replayed as a plugin.

The interesting question is not whether this works; it is known to. It is **which architecture** should do it. Dilated causal convolutions in the style of WaveNet have been the standard since around 2019, while transformers have displaced convolutional baselines almost everywhere else in machine learning.

So we built both, under one shared pipeline, and compared them at matched parameter count and matched receptive field. Everything except the architecture is held fixed: the same loss, the same optimiser and schedule, the same seed, the same data.

§ 2 Data

We use the IDMT-SMT-GUITAR_V2 corpus, whose fourth subset contains long direct recordings of two electric guitars: an Ibanez 2820 and a Career SG, roughly an hour each. The clips are concatenated into one continuous recording per guitar using the OpenAmp preparation script, giving 42 and 44 minutes of usable audio at 44.1 kHz.

These recordings are the **input**. The targets are generated by running them through deterministic DSP amplifiers built with **pedalboard**, which makes the ground truth exact and reproducible rather than dependent on a microphone placement. Four targets are produced, chosen so that they isolate different difficulties:

target	chain	character
clean	empty	identity – a control
preamp	gate, high-pass, 35 dB distortion, EQ	nonlinear, memory of a few ms
cabinet	impulse-response convolution	purely linear, long tail
metal	preamp followed by cabinet	both together

The split is deliberate: **preamp** isolates nonlinearity from long memory, and **cabinet** isolates long memory from nonlinearity, so a model that struggles with one and not the other can be diagnosed.

Training uses the Ibanez; validation uses the Career SG. Holding out a **different guitar** is a real generalisation test, where a random split of one recording would only measure interpolation within the same take and tone.

Each example is one second of target audio (44,032 samples) preceded by half a second of dry lead-in that the model may read but is not scored on, so that no prediction is made from an all-zero history. Segments whose target is quieter than -40 dBFS are dropped: every loss here is relative to target energy, so a near-silent target divides by almost nothing. The validation guitar yields 2,596 segments.

§ 3 Model

Both architectures are causal, operate on raw samples, and predict the residual – the difference the amplifier makes – rather than the signal itself.

§3.1 WaveNet

The baseline. A stack of dilated causal convolutions with gated activations, where the dilation doubles at each layer and the stack repeats, so the receptive field grows exponentially with depth while the parameter count grows linearly. Each layer contributes to a skip sum, which two 1×1 convolutions project down to the output.

Because every layer is a convolution over samples, the model works at the full sample rate from input to output.

§3.2 AmpFormer

The experimental arm: a block-causal transformer.

Attention cannot run at the sample rate. One second of audio is 44,100 positions, and comparing each against every other is about 1.9×10^9 pairs – roughly 7.8 GB of scores per head per second in `float32`. So samples are first grouped into chunks of 64, bringing a second down to 689 positions. The grouping is a convolution whose stride equals its kernel size, which is exactly chunking plus a learned projection of each chunk.

Three further decisions follow from that:

- **Bounded causal attention.** A sliding mask lets chunk i attend to chunks $i - w + 1$ through i , so the model never sees the future and never reaches further back than the window allows. Reach compounds over depth, giving a receptive field of $(\text{layers} \cdot (w - 1) + 1) \cdot \text{chunk samples}$. The window is quoted in chunks precisely so that it can be set to match WaveNet’s receptive field exactly.
- **Rotary position embeddings.** A rotary dot product depends only on the **difference** between two positions. That is what allows a streaming decoder to keep a key/value cache, resume from wherever it stopped, and still reproduce the attention it was trained with.
- **A zero-initialised output.** The model begins as an exact passthrough and has to learn only the amplifier’s contribution, never the guitar signal itself.

The cost of chunking is that the model emits one vector per 64 samples and must reconstruct those 64 samples from it. Section 4 is largely about what that costs.

§3.3 Matched budget

model	parameters	context	latency
WaveNet 46 ms	205,505	46.4 ms	0.02 ms
AmpFormer 48 ms	207,297	47.9 ms	1.45 ms

WaveNet 186 ms	243,265	185.7 ms	0.02 ms
AmpFormer 187 ms	207,297	187.2 ms	1.45 ms

At the short context the two agree to within 1% on parameters. At the long context WaveNet carries 17% **more** parameters, so any advantage there is the baseline's, not the transformer's.

Latency differs structurally and cannot be tuned away: WaveNet emits a sample as soon as it has one, while AmpFormer cannot emit a chunk until all 64 of its samples have arrived. Both remain far below what a player would notice.

§ 4 Evaluation

§4.1 The metric

Plain mean squared error is a poor fit. It is dominated by the loud fundamental, while the character of an amplifier lives in the comparatively quiet harmonics it generates, and it counts a loudly played passage for far more than a quiet one.

We report error-to-signal ratio instead, normalised per segment:

$$\text{ESR} = \frac{1}{N} \sum_{i=1}^N \frac{\text{mean}((y_i - \hat{y}_i)^2)}{\text{mean}(y_i^2) + \varepsilon}$$

Dividing by each segment’s own energy makes a quietly played second count as much as a loud one. The training objective adds a pre-emphasised ESR term to weight the harmonics, a multi-resolution STFT term, and a DC term, but ESR is the number reported because it is the one comparable across architectures.

Evaluation runs a checkpoint over all 2,596 validation segments in a single pass and averages. Each model sees the lead-in but is scored only on the tail.

§4.2 Results

target	context	WaveNet	AmpFormer	gap
cabinet	46 ms	0.00010	0.00967	97×
cabinet	186 ms	0.00004	0.01042	260×
preamp	46 ms	0.00091	0.00695	8×
preamp	186 ms	0.00022	0.00770	35×
metal	46 ms	0.00622	0.01263	2×
metal	186 ms	0.00075	0.00565	8×

`clean` is omitted: it is an identity mapping, both models solve it trivially given the residual connection, and the reported figure hits the floor imposed by ε . It separates nothing, which is what a control is for.

WaveNet wins everywhere. Two patterns in [how](#) it wins are more informative than the fact itself.

§4.3 Nonlinearity is cheaper than long memory

Comparing the two isolated targets: `preamp` is nonlinear with a few milliseconds of memory, `cabinet` is perfectly linear with a long tail. Against WaveNet, AmpFormer is roughly an order of magnitude further behind on the cabinet than on the preamp.

This is what the chunk bottleneck predicts. Reproducing a long impulse response requires sample-accurate phase, and phase cannot be recovered from a per-chunk summary. A saturating transfer curve is far more forgiving, because the residual connection already supplies the waveform the curve is applied to.

metal – which contains the same cabinet convolution – nonetheless shows the **smallest** gap. The likely reason is that ESR normalises by target energy, and 35 dB of distortion dominates that energy, so the same absolute filtering error reads as a much smaller relative one. We have not verified this.

§4.4 More context does not help the transformer

Going from roughly 46 ms of context to roughly 186 ms:

target	WaveNet	AmpFormer
cabinet	2.5× better	0.9× – worse
preamp	4.1× better	0.9× – worse
metal	8.3× better	2.2× better

A model starved of context benefits from more of it. AmpFormer does not; on two of three targets it gets worse. Whatever limits it is not reach, which leaves resolution within the chunk. Extra context cannot restore detail that the bottleneck has already discarded.

§4.5 How the numbers should be read

Each figure is the **minimum** validation ESR across all epochs, because the best epoch is the one kept and early stopping watches the same quantity. Selecting the best of several hundred noisy measurements is optimistic, so these are upper bounds on quality rather than unbiased estimates. The validation set is large and the gaps range from twofold to over two hundredfold, far outside that bias, so the comparison stands even though the absolute values are flattering.

Both architectures trained under identical early stopping, so neither was cut off by a budget; on the two targets where the gap matters, AmpFormer in fact trained longer.

§ 5 Conclusion

Under a controlled comparison at matched parameter count and matched receptive field, the dilated convolution beats the block-causal transformer at every target and every context length, by factors from two to two hundred and sixty.

The failure is specific rather than general. AmpFormer falls furthest behind on the purely linear, long-memory target and least on the distorted one, and it does not improve when given four times the context. Both observations point at the same cause: compressing 64 samples into a single vector and reconstructing them from it discards resolution that the task needs and that no amount of additional reach restores.

That suggests the fix rather than abandoning the idea. A transformer could predict **filter coefficients** per chunk instead of samples per chunk, leaving the filtering itself at the sample rate. Attention would stay cheap while the output regained sample resolution.

What we have not done bounds the claim. There is no ablation over chunk size, which is the direct test of the diagnosis above and the obvious next experiment; the coefficient-predicting variant is a proposal and nothing more; each configuration was trained with one seed; and the validation guitar doubles as the model-selection set, so a clean held-out test would need a third instrument that this corpus does not offer at comparable length.

What the project does establish is a fair comparison rather than a claim, a shared pipeline in which either architecture can be dropped, and a specific, testable reason why the transformer loses.